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Alpine Linux Persistence and Storage Summit 2025

Detecting Supply Chain Attacks from the Filesystem Level with eBPF, fanotify and others

Tobias Specht
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Agenda

- **Background**
- Supply chain attacks and the XZ backdoor
- Previous work
- **Linux Tracing APIs**
- **Requirements**
- **API Experience**
- **Solution**
- **POC**
- **Future work**

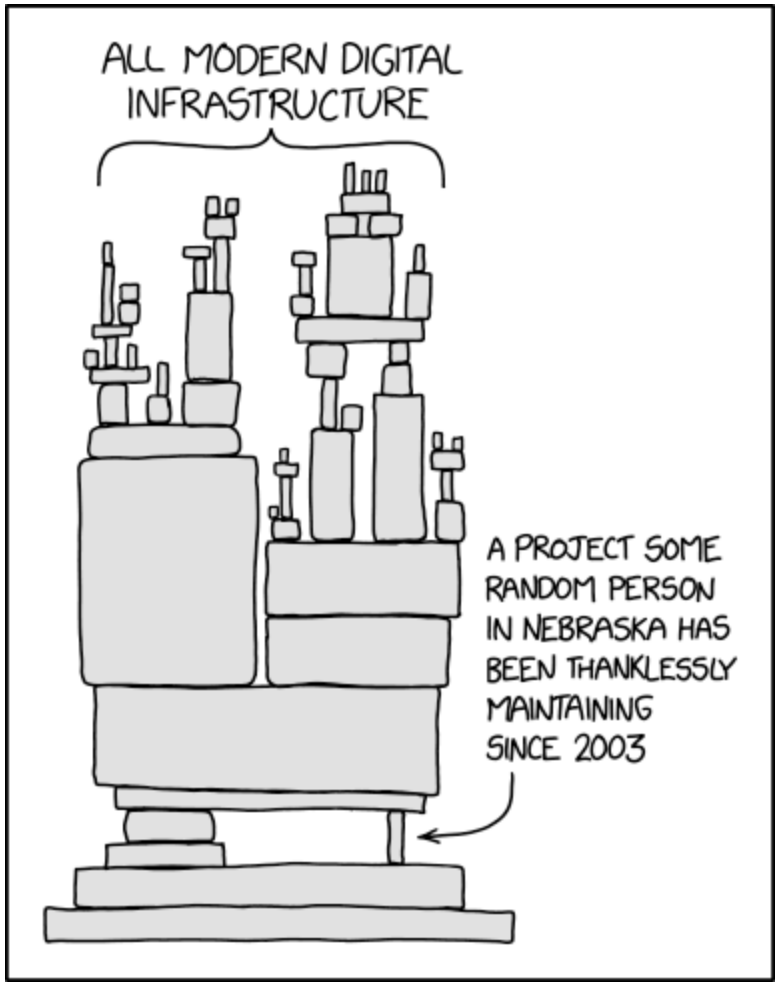
Whoami

Tobias Specht

- **Cybersecurity researcher based in Germany**
- **Working at the Fraunhofer Institute for Applied and Integrated Security (AISEC)**
- **Research focus on**
- Static Code Analysis
- Offensive Security
- Automotive and Embedded Domain
- **https://github.com/peckto**
- **Matrix: @peckto:tchncs.de**

Background

Background
Supply Chain Attacks

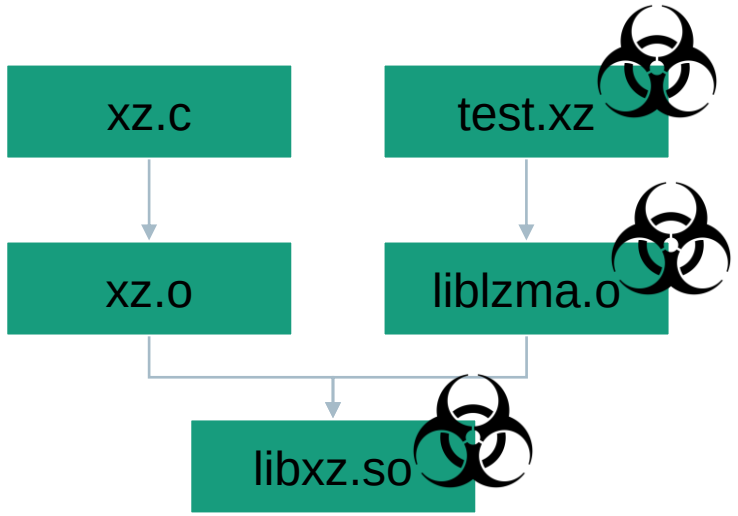


https://xkcd.com/2347

Background

XZ Backdoor CVE-2024-3094 (v5.6.1)

- Target of the attack was the openssh server, to gain remote code execution
- Infiltrating the indirect dependency xz
- Attack on xz was only detected by coincidence, it’s unknown how many undetected attacks exist (or have existed)
- Supply chain attack on xz was utilizing the build system to hide and inject its malicious code



Supply Graph
FOSDEM 2025

Method

- Trace execv to capture all program invocations (inspired by compile_commands.json)
- Reconstruct dataflow based on cli arguments (e.g. gcc source.c –o exe, cp source dest, …)

Limitations

- Need to parse cli
- Not all dataflows are visible (e.g. IPC)

Idea:

1. Monitor on filesystem level
2. Somehow trace pipes to capture relevant IPC

Supply Graph

FOSDEM 2025

README Apache-2.0 license

Analyze build process

In the docker container, the following Debian packet builds are included:

- xz-5.6.1 (CVE-2024-3094)
- xz-5.6.2
- openssh-9.2p1
- openssl-3.0.15

Run the analysis:

```
docker run --rm -it supply-graph
analyze-build-graph xz-5.6.1
```

Identified anomalies in the supply graph are displayed at the end of the log:

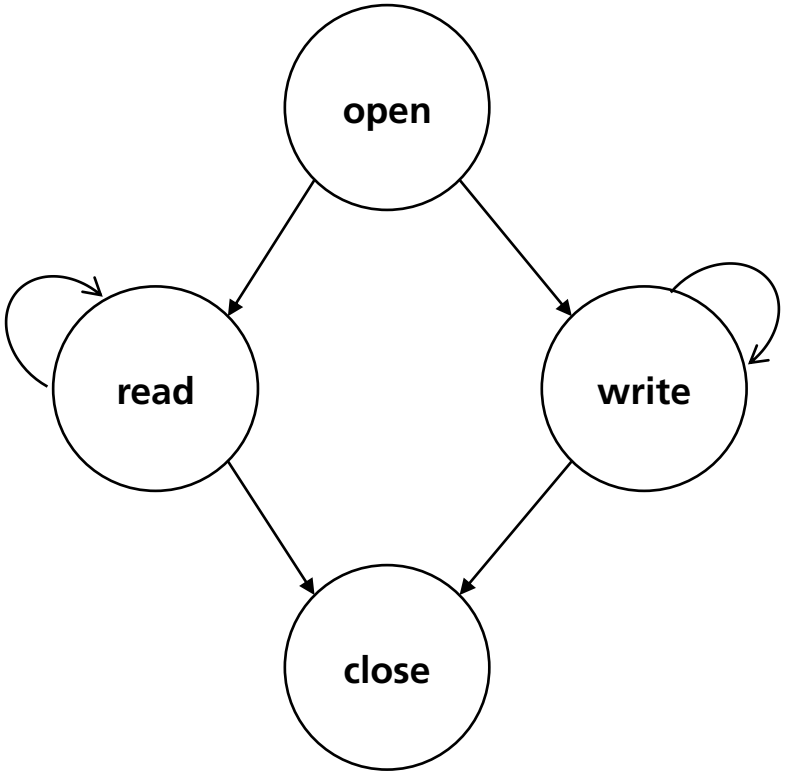
```
[...]
Root files not part of upstream:
* /data/xz-5.6.1/xz-utils-5.6.1/debian/normal-build/src/liblzma/liblzma_la-crc64-fast.o
Binary files without corresponding source code:
* /data/xz-5.6.1/xz-utils-5.6.1/debian/normal-build/src/liblzma/liblzma_la-crc64-fast.o
```

https://github.com/Fraunhofer-AISEC/supply-graph

Tracing on filesystem level

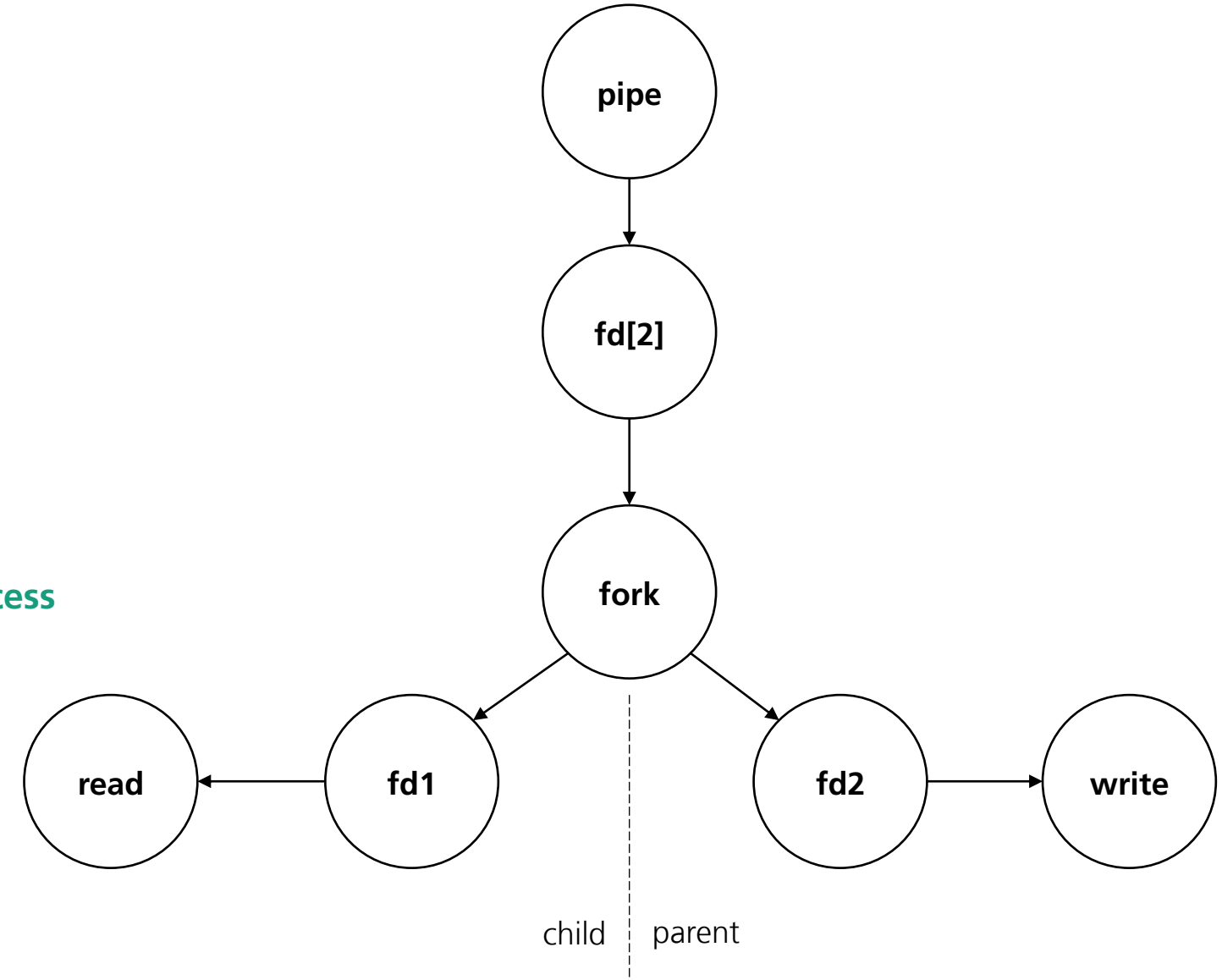
Tracking file access

- Operations: open, read, write, close
- File descriptor is not stateless



Tracking pipes

- **Operations**
- File: pipe, read, write, close
- Process: fork | clone
- **File descriptor is not stateless**
- **Need to track source and sink process**



Linux Tracing APIs

API	Space	Interface	Sync/Async
ptrace	User	Syscals	Sync
LD_PRELOAD	User	Library calls	Sync
eBPF	Kernel	Syscals	Async
Inotify	User	File operations	Async
fanotify	Root	File operations	Sync

Requirements

Requirement	ptrace	LD_PRELOAD	eBPF	Inotify	fanotify
Transparent	x	x	x	x	x
Stealth	-	-	x	x	x
Track file OP	x	x	x	x	x
Track pipe OP	x	x	x	-	-
Container	x	x	~	x	x
Complexity	low	medium	medium	medium	medium

API Experience

API Experience
ptrace (strace)

```
[pid 36398] openat(AT_FDCWD, "/lib64/liblzma.so.5", O_RDONLY|O_CLOEXEC) = 3
[pid 36399] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[pid 36399] close(3)                    = 0
[pid 36399] openat(AT_FDCWD, "/lib64/libtinfo.so.6", O_RDONLY|O_CLOEXEC <unfinished ...>
[pid 36398] close(3 <unfinished ...>
[pid 36399] <... openat resumed>)       = 3
[pid 36398] <... close resumed>)        = 0
```

- **Pro**
- Ready to run tool strace, with filter options
- Capture only events from target process(es)

- **Con**
- Output not machine readable (no json format)
- Difficult to align operations (unfinished…, resumed)
- Hooks would need custom implementation

API Experience
LD_PRELOAD

- **Pro**
- Full control
- Example projects to build up on
- Capture only events from target process(es)
- **Con**
- Not verry elegant
- Much manual effort
- Living inside the target process

API Experience
eBPF (bpftrace)

- **Pro**
- Transparent and stealth for the process
- Very performant
- High level language to describe trace points (bpftrace)
- **Con**
- Captures all system events! Need to filter wisely
- Async event queue and no guarantee for processing order (close might be processed bevor open)
- No function hooking or influence on the user process
- No access to userland (e.g. no access to file content, only to kernel structs)

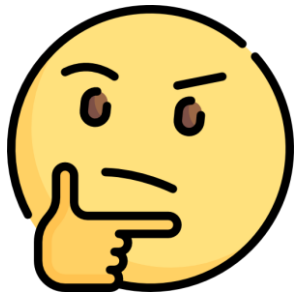
API Experience
fanotify

- **Pro**
- Permission events are sync
- Access to file content

- **Con**
- Captures all system events! Need to filter wisely
- Not all file events are sync (FAN_ACCESS_PERM, FAN_OPEN_PERM, FAN_OPEN_EXEC_PERM)
- No alignment between pre and post events

API Experience
Summary

- **ptrace** ❌
- **LD_PRELOAD** ❌
- **eBPF** 〰
- **fanotify** 〰



Open problems:

1. Stateless way to track pipes
2. Trace all file access in sync and capture content, before and after write

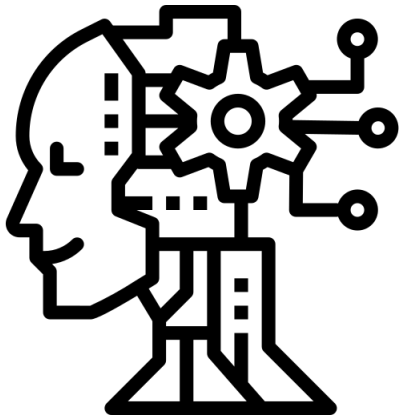
Solution

Solution
Part 1 – pipe

Me

This is my problem

- Use fstat(fd) and take st_ino
  That uniquely identifies the underlying pipe object while it exists
- Or read /proc/fd/ (a symlink like "pipe:[12345]");
  the number is the pipe's inode



AI

Solution

Part 2 – file access

Me

This is my problem

Sounds like you need to write your own filesystem

You can use FUSE fs, they have examples

Colleague

Solution

- **FUSE fs**
- Build up on libfuse passthrough example
- Convert fd to unique counter
- Track file operations
- Hash file content
- **Bpftrace**
- Track pipes based on inode number

Setup

- **VM with Debian 12**
- **FUSE mount / as /mnt**
- **Bind mound special file systems (e.g. /dev, /proc, …) on /mnt**
- **Chroot /mnt**
- bpftrace <trace_pipe.bpftrace> -c <target command>
- **Cleanup mounts**
- **Create data flow graph in KuzuDB**
- **Write graph queries in Cypher**

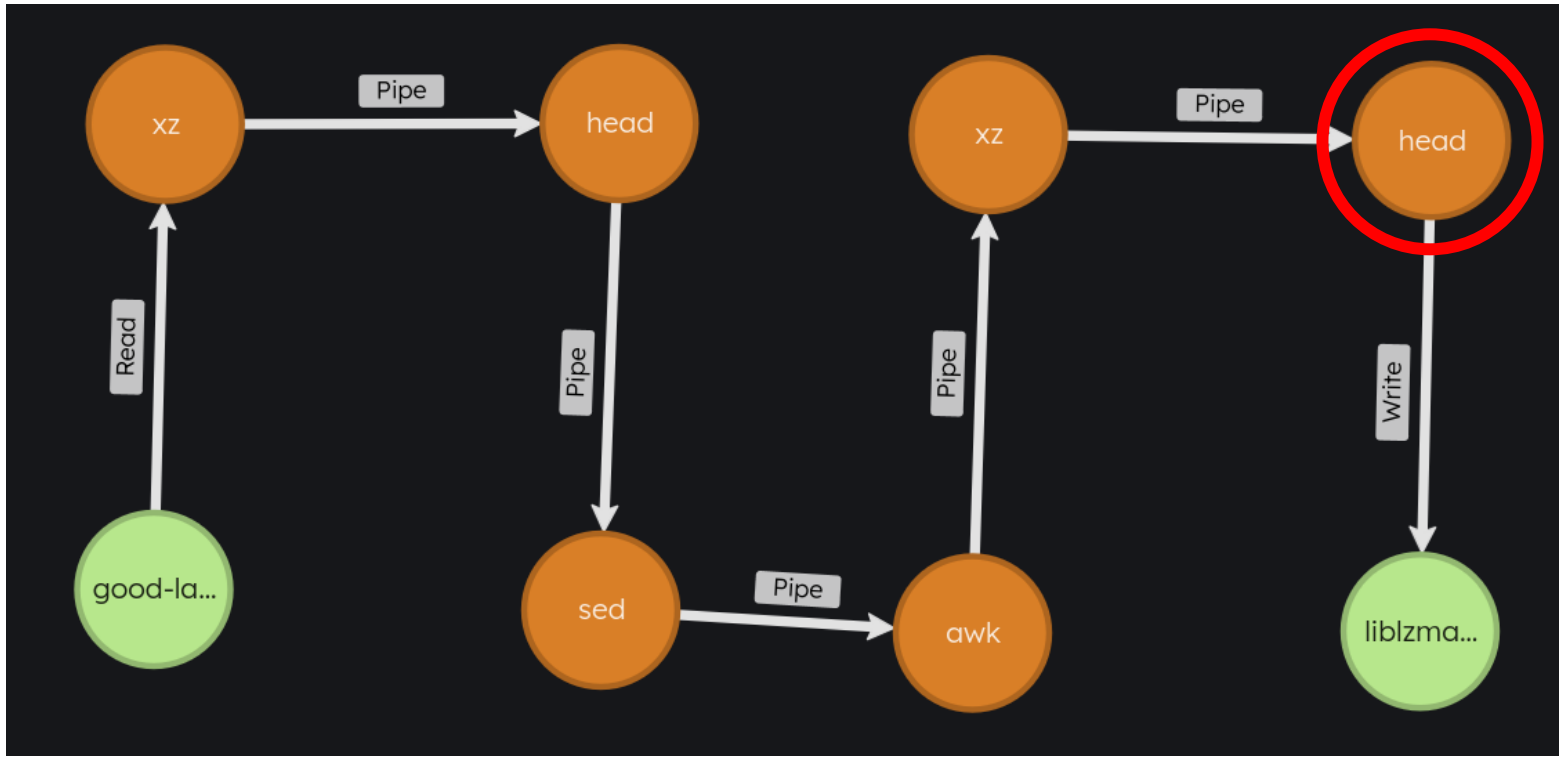
Limitations

- Currently no Docker support
- No support for other IPC methods (e.g. socket, queue, …)

POC

CVE-2024-3094

```
match (l:File {name: "good-large_compressed.lzma"})-[a* SHORTEST 1..30]->(f:File {path: "liblzma_la-crc64-fast.o"})
return l, a, f
```



POC

CVE-2024-3094

```
MATCH (p:Process)-[w:Write]->(f:File {type: ".o"})
where
p.name <> "strip" and
p.name <> "cp" and
p.name <> "as" and
p.name <> "ld" and
p.name <> "x86_64-linux-gnu-as" and
p.name <> "x86_64-linux-gnu-ld.gold"
RETURN f, p
```

Future Work

- Write more queries to detect anomalies
- Evaluate approach with more packages
- Integrate into package build CI

Thanks

- @rumpelsepp
- @dxld

Acknowledgements

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https://www.forschung-it-sicherheit-kommunikationssysteme.de/projekte/alpaka

Contact

Tobias Specht
Group Embedded Software Analysis
Department Product Protection & Industrial Security
Phone +49 89 32299 86 187
tobias.specht@aisec.fraunhofer.de

Fraunhofer Institute for Applied and Integrated Security AISEC
Lichtenbergstr. 11
85748 Garching
Germany

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